

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 2 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. What is the formula to calculate the total number of matches in a Double League tournament? (A) $N - 1$ (B) $\frac{N(N-1)}{2}$ (C) $N(N - 1)$ (D) $\frac{N+1}{2}$ *Unit 1: Management of Sporting Events | Tag: Very High***

Q2. The onset or beginning of the menstrual period in young females is medically termed as: (A) Amenorrhea (B) Menopause (C) Dysmenorrhea (D) Menarche *Unit 2: Children & Women in Sports | Tag: Very High***

Q3. Which of the following yoga asanas is famously translated as the "Cobra Pose"? (A) Gomukhasana (B) Bhujangasana (C) Halasana (D) Vajrasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q4. The expanded form of ADHD, a common behavioral disorder in children, is: (A) Attention Deficit Hyperactivity Disorder (B) Attention Defect Hyperactive Disability (C) Automatic Deficit Hyperactive Disorder (D) Attention Defiant Hyperactivity Disorder *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q5. Which of the following is classified as a Macro-mineral essential for bone density and muscle contraction? (A) Iron (B) Iodine (C) Calcium (D) Copper *Unit 5: Sports & Nutrition | Tag: Very High***

Q6. In the Partial Curl-Up test used for assessing abdominal strength, what is the standard distance between the two parallel lines on the testing mat? (A) 4 inches (B) 6 inches (C) 8 inches (D) 10 inches *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q7. The short-term physiological effect of intense exercise that causes muscle fatigue and a burning sensation is the accumulation of: (A) Glycogen (B) Lactic acid (C) ATP (D) Oxygen *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q8. Newton's Second Law of Motion is also universally known as the: (A) Law of Inertia (B) Law of Action and Reaction (C) Law of Momentum / Acceleration (D) Law of Gravitation *Unit 8: Biomechanics and Sports | Tag: Very High***

Q9. The "Staircase Method" and "Tabular Method" are procedures used to draw fixtures for which type of tournament? (A) Knock-out Tournament (B) League Tournament (C) Challenge Tournament (D) Combination Tournament *Unit 1: Management of Sporting Events | Tag: High***

Q10. The postural deformity characterized by an exaggerated inward curvature of the lumbar spine (lower back) is called: (A) Kyphosis (B) Lordosis (C) Scoliosis (D) Flat Foot *Unit 2: Children & Women in Sports | Tag: Very High***

Q11. Who among the following is recognized as the founder of the Paralympic movement? (A) Baron Pierre de Coubertin (B) Eunice Kennedy Shriver (C) Sir Ludwig Guttmann (D) John Fontanella *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q12. A person with a calculated Body Mass Index (BMI) of 27.5 falls into which WHO category? (A) Underweight (B) Normal weight (C) Overweight (D) Obese Class I *Unit 5: Sports & Nutrition | Tag: High***

Q13. The Rockport One-Mile Walk Test is primarily administered to estimate an individual's: (A) Flexibility (B) Muscular Endurance (C) Agility (D) $\dot{V}O_2$ Max (Cardiovascular fitness) *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q14. The scraping or rubbing away of the upper layer of the epidermis due to friction with a rough surface is classified as a/an: (A) Contusion (B) Sprain (C) Abrasion (D) Laceration *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q15. A bicep curl exercise is a classic anatomical example of which class of lever in the human body? (A) Class I Lever (B) Class II Lever (C) Class III Lever (D) Class IV Lever *Unit 8: Biomechanics and Sports | Tag: High***

Q16. Assertion (A): A sprinter resting in the starting blocks will remain completely at rest until muscular force initiates the sprint. **Reason (R):** According to Newton's First Law of Motion (Inertia), an object at rest stays at rest unless acted upon by an external resultant force. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | Tag: Very High***

Q17. Assertion (A): The Plate Tapping Test measures the speed and coordination of the upper limbs. **Reason (R):** The Plate Tapping Test is a mandated component of the SAI Khelo India fitness battery for the older age group of 9-18 years. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 6: Test and Measurement in Sports | Tag: High***

Q18. Assertion (A): Vitamins and minerals are biologically categorized as protective foods. **Reason (R):** Vitamins and minerals provide extremely high amounts of caloric energy required for intense sports performance. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | Tag: Very High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Mention any two pre-event responsibilities of the Marketing Committee during the organization of a sports tournament. *Unit 1: Management of Sporting Events | Tag: High***

- Q20.** Suggest any two corrective physical activities or yoga asanas for the postural deformity known as 'Round Shoulders'. *Unit 2: Children & Women in Sports | Tag: Very High***
- Q21.** Differentiate strictly between the target participants of the Special Olympics and the Paralympics. *Unit 4: Physical Education & Sports for CWSN | Tag: High***
- Q22.** State any two common myths related to food and dieting. Give the actual fact for one of them. *Unit 5: Sports & Nutrition | Tag: Very High***
- Q23.** What is the main purpose of the '8-foot up and go test' in the Senior Citizen Fitness Test? List the equipment required for it. *Unit 6: Test and Measurement in Sports | Tag: Very High***
- Q24.** What does the acronym PRICE stand for in the immediate first-aid management of soft tissue injuries? *Unit 7: Physiology & Injuries in Sport | Tag: Very High***
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SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

- Q25.** What is a Consolation Tournament? Briefly explain its two main types. *Unit 1: Management of Sporting Events | Tag: Medium***
- Q26.** Elaborate on the Physical Exercise Guidelines recommended by the World Health Organization (WHO) for the age group of 5-17 years. *Unit 2: Children & Women in Sports | Tag: Very High***
- Q27.** Make a list of any three Yoga Asanas recommended for the prevention of Back Pain. Explain the step-by-step procedure of any one of them. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: High***
- Q28.** Discuss the concept of 'Divisioning' in disability sports. Why is it an essential process? *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***
- Q29.** Regular exercise brings profound changes to the human body. Explain any three physiological changes that occur specifically due to the natural process of aging. *Unit 7: Physiology & Injuries in Sport | Tag: High***
- Q30.** Define 'Projectile' in sports. Explain any two aerodynamic factors (other than gravity) that affect the trajectory of a projectile. *Unit 8: Biomechanics and Sports | Tag: Very High***
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SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

- Q31.** The Physical Education teacher of XYZ Public School has been tasked with organizing an Inter-House Basketball competition on a Knock-out basis. A total of 21 teams have submitted their entries. He sits down to calculate the mathematical aspects of the fixture to ensure a fair draw. **(i)** What will be the total number of matches played in this tournament? **(ii)** Calculate the total number of byes that need to be given in the first round. **(iii)** How many

teams will be placed in the upper half of the fixture? **(iv)** How many total rounds will be played in this tournament? **OR** Which half (upper or lower) will receive the extra bye, and why? *Unit 1: Management of Sporting Events | Tag: Very High***

Q32. The Sports Authority of India (SAI) mandates the Khelo India Fitness Test to track the motor and physical development of school children. A school conducts this fitness battery for its Class 6 to 12 students (Age group 9-18 years) on their annual sports day. **(i)** Which specific test from this battery should the teachers administer to measure the Cardiovascular Endurance of the students? **(ii)** Which test should be conducted to evaluate their Abdominal/Core Strength? **(iii)** To evaluate explosive acceleration and speed, which sprint test is used? **(iv)** How is flexibility measured in this specific age category? **OR** State the mathematical formula used to calculate a student's Body Mass Index (BMI). *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q33. Biomechanics plays a crucial role in analyzing sports techniques. During a golf coaching camp, the coach explains that the golf ball remains perfectly stationary on the tee until the club strikes it. Furthermore, the harder the golfer swings the club, the faster and further the ball travels across the fairway. **(i)** The ball remaining at rest until struck by the club is a practical application of which of Newton's Laws? **(ii)** When the golfer bends their elbow to swing, the bicep acts as the effort in the middle. Which class of lever is this? **(iii)** The parabolic flight path followed by the golf ball in the air is scientifically termed as its __. **(iv)** Name the invisible downward force that pulls the golf ball back to the ground. **OR** Newton's Third Law of Motion is also known as the Law of __. *Unit 8: Biomechanics and Sports | Tag: Very High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. Draw a neat knock-out fixture of 19 teams. Clearly show all the necessary calculations including the number of matches, total byes, byes in the upper and lower halves, and the number of rounds. *Unit 1: Management of Sporting Events | Tag: Very High***

Q35. What is Obesity? Enlist any four Yoga Asanas recommended for its prevention. Thoroughly describe the procedure, two benefits, and two contraindications of *Bhujangasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q36. Define a 'Balanced Diet'. Elaborate comprehensively on the functions and specific physiological importance of any four Macro-Nutrients (e.g., Carbohydrates, Proteins, Fats, Water) in an athlete's regular diet. *Unit 5: Sports & Nutrition | Tag: High***

Q37. State Newton's Three Laws of Motion. Thoroughly explain the practical application of each of these three laws in sports events, providing at least one distinct real-world sports example for each law. *Unit 8: Biomechanics and Sports | Tag: Very High***

Answer Key

Q1. (C) $N(N - 1)$ **Q2.** (D) Menarche **Q3.** (B) Bhujangasana **Q4.** (A) Attention Deficit Hyperactivity Disorder **Q5.** (C) Calcium **Q6.** (B) 6 inches **Q7.** (B) Lactic acid **Q8.** (C) Law of Momentum / Acceleration **Q9.** (B) League Tournament **Q10.** (B) Lordosis **Q11.** (C) Sir Ludwig Guttmann **Q12.** (C) Overweight **Q13.** (D) $\dot{V}O_2$ Max (Cardiovascular fitness) **Q14.** (C) Abrasion **Q15.** (C) Class III Lever **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (C) (A) is true, but (R) is false. (Reason is false because Plate Tapping is for 5-8 years, not 9-18 years). **Q18.** (C) (A) is true, but (R) is false. (Reason is false because vitamins/minerals provide no caloric energy).

Q19. 1. Preparing a strategy for arranging sponsorships and securing funds. 2. Managing media publicity, press releases, and campaigns to promote the upcoming event. **Q20.** 1. *Chakrasana* (Wheel pose) or *Dhanurasana* (Bow pose) to open the chest. 2. Holding a horizontal bar and hanging for some time to stretch the pectoral muscles. **Q21.** The **Special Olympics** is strictly for children and adults with intellectual/cognitive disabilities (e.g., Autism, Down syndrome). The **Paralympics** is an elite sports competition for athletes with physical, visual, and intellectual impairments. **Q22. Myth 1:** Carbohydrates make you fat. (Fact: Excess calories make you fat; complex carbs are essential for sustained energy). **Myth 2:** Skipping breakfast helps in quick weight loss. **Q23. Purpose:** To assess agility and dynamic balance in senior citizens, which is crucial for tasks like getting off a bus. **Equipment:** A folding chair, a stopwatch, a cone marker, and a measuring tape. **Q24. P** = Protection (from further injury), **R** = Rest (stop playing), **I** = Ice (apply cold pack to reduce swelling), **C** = Compression (wrap with a bandage), **E** = Elevation (keep injured part above heart level).

Q25. A Consolation Tournament is organized to provide a second chance to teams that get eliminated in the early rounds of a knock-out tournament. **Type 1:** Teams defeated in the *first round only* play again. **Type 2:** Teams defeated in *all regular preliminary rounds* (up to quarter-finals) play again to decide subsidiary rankings. **Q26.** WHO Guidelines (5-17 years): 1. Accumulate at least 60 minutes of moderate-to-vigorous intensity aerobic physical activity daily. 2. Amounts greater than 60 mins yield additional health benefits. 3. Incorporate vigorous-intensity activities that strengthen muscle and bone at least 3 times a week. 4. Limit sedentary screen time. **Q27. Asanas for Back Pain:** Tadasana, Ardh Chakrasana, Bhujangasana, Vakrasana. **Procedure for Tadasana:** Stand straight with feet together. Interlock fingers, turn palms upwards. Inhale and stretch arms up while raising heels off the ground. Balance on toes, stretch the whole body upwards. Exhale and bring heels down. **Q28. Divisioning** is the process of grouping athletes in disability sports (like Special Olympics) based on their age, gender, and specifically their *ability level*. It is essential because it ensures fair and equitable competition, giving every athlete an equal chance to excel and win, fulfilling the philosophy of inclusion. **Q29.** 1. **Bone Density:** Bones lose minerals and become brittle (osteoporosis), increasing fracture risks. 2. **Muscle Mass:** Muscle size and strength decrease (sarcopenia), slowing down metabolism. 3. **Cardiovascular:** The heart muscle becomes less efficient, reducing maximum heart rate and $\dot{V}O_2$ max, leading to quicker

fatigue. **Q30. Projectile:** Any object thrown into the air that moves under the influence of gravity (e.g., a thrown javelin or kicked football). **Aerodynamic Factors:** 1. *Air Resistance:* Friction from the air slows down the horizontal velocity. 2. *Spin/Magnus Effect:* A spinning ball creates air pressure differences, causing it to lift or drop faster than a non-spinning ball.

Q31. (i) Total Matches = $N - 1 = 21 - 1 = 20$ matches. (ii) Byes = Next power of 2 minus $N = 32 - 21 = 11$ byes. (iii) Upper half teams = $\frac{N+1}{2} = \frac{22}{2} = 11$ teams. (iv) Rounds = Next power of 2 is 2^5 , so 5 rounds. **OR** Lower half gets the extra bye, because the formula for byes in the lower half is $\frac{N_b+1}{2}$ ($\frac{11+1}{2} = 6$ byes in lower, 5 in upper). **Q32.** (i) 600m Run/Walk test. (ii) Partial Curl-up test (for 30 seconds). (iii) 50m Dash (Standing Start). (iv) Sit and Reach Test. **OR** BMI = $\frac{\text{Weight (in kg)}}{\text{Height (in meters)}^2}$. **Q33.** (i) Newton's First Law (Law of Inertia). (ii) Class III Lever (Effort in the middle). (iii) Trajectory (or Parabola). (iv) Gravity. **OR** Law of Action and Reaction.

Q34. Calculations: Total Teams (N) = 19. Matches = $19 - 1 = 18$. Next power of 2 = 32. Total Byes = $32 - 19 = 13$. Teams in Upper Half = $\frac{19+1}{2} = 10$. Teams in Lower Half = $\frac{19-1}{2} = 9$. Byes in Upper Half = $\frac{13-1}{2} = 6$. Byes in Lower Half = $\frac{13+1}{2} = 7$. Number of Rounds = 5 (since $2^5 = 32$). (Evaluator Note: Draw a clear bracket showing 19 teams. Upper half has 10 teams, lower has 9. Place 6 byes in upper, 7 in lower following the standard bottom-top-bottom-top allocation. Connect matches across 5 rounds to declare one winner). **Q35. Obesity** is an abnormal accumulation of excess body fat posing health risks ($BMI > 30$). **Asanas:** Vajrasana, Tadasana, Trikonasana, Bhujangasana. **Bhujangasana Procedure:** Lie prone on the stomach, legs together. Place hands on the floor beside the shoulders. Inhale and slowly lift the head, chest, and abdomen, arching the back using back muscles. Hold, exhale, and return. **Benefits:** Tones abdominal organs to aid weight loss; improves spinal flexibility.

Contraindications: Avoid if suffering from a hernia, recent abdominal surgery, or severe back injuries/pregnancy. **Q36. A Balanced Diet** contains all essential macro and micro-nutrients in correct proportions required for optimum body function and athletic performance. 1. **Carbohydrates:** The primary and fastest source of energy (4 kcal/g); stored as glycogen in muscles for endurance and high-intensity work. 2. **Proteins:** The "building blocks" (4 kcal/g); essential for repairing micro-tears in muscle tissues post-workout and promoting muscle hypertrophy. 3. **Fats:** A concentrated energy source (9 kcal/g); cushions vital internal organs and helps transport fat-soluble vitamins (A, D, E, K). 4. **Water:** Though providing 0 calories, it makes up ~60% of body weight, regulates core temperature via sweating, and transports nutrients to cells. **Q37. 1. Law of Inertia:** An object remains at rest or in uniform motion unless acted upon by an external force. *Example:* A soccer ball remains stationary on the penalty spot until a player applies muscular force to kick it. **2. Law of Acceleration ($F = ma$):** The acceleration of an object is directly proportional to the force applied and inversely proportional to its mass. *Example:* In Shot Put, an athlete must exert significantly more explosive force to accelerate the heavy iron ball compared to throwing a lighter baseball. **3. Law of Action and Reaction:** For every action, there is an equal and opposite reaction. *Example:* A sprinter pushes backward and downward against the starting blocks (Action), and the blocks push the sprinter forward and upward with equal force (Reaction) to start the race.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.